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Part III: Workout Questions [5 pts]

1. Use truth table method to determine the validity of the argument: [2 pts]

   $p \Rightarrow q, \neg r \Rightarrow \neg q \vdash \neg r \Rightarrow \neg p$

						P	P	C
p	q	r	$\neg r$	$\neg q$	$\neg p$	$p \Rightarrow q$	$\neg r \Rightarrow \neg q$	$\neg r \Rightarrow \neg p$
T	T	T	F	F	F	T	T	T
T	T	F	T	F	F	T	F	F
T	F	T	F	T	F	F	T	T
T	F	F	T	T	F	F	T	F
F	T	T	F	F	T	T	T	T
F	T	F	T	F	T	T	F	T
F	F	T	F	T	T	T	T	T
F	F	F	T	T	T	T	T	T

2

So It is Valid b/c when the premises are True the conclusion also true

2. *Find the square roots of the complex number* $z = 9i$ [3 pts]

$r = 9$

$\theta = \tan^{-1}\left(\frac{9}{0}\right) = \text{undefined}$

$\theta = \pi/2$

$z = re^{i\theta}$

$z_0 = r^{1/n} e^{i\left(\frac{\theta}{n} + \frac{2\pi k}{n}\right)}$

$z_0 = 9^{1/2} e^{i\left(\frac{\pi/2}{2} + \frac{2\pi k}{2}\right)}$ $k = n-1$

$k = 0, 1$

when $k = 0$

$z_0 = 9^{1/2} e^{i(\pi/4)}$

$= 3e^{i(\pi/4)}$

$z_0 = r(\cos\theta + i\sin\theta)$

$= 3(\cos(\pi/4) + i\sin(\pi/4))$

$\boxed{z_0 = \frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}i}$

when $k = 1$

$z_0 = 9^{1/2} e^{i\left(\frac{\pi}{4} + \pi\right)}$

$= 3e^{i(5/4\pi)}$

$z_0 = r(\cos\theta + i\sin\theta)$

$= 3\left(\cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}\right)$

3

$\boxed{z_0 = \frac{-3\sqrt{2}}{2} - \frac{3\sqrt{2}}{2}i}$

9.5/15

ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY

College of ------CNSS-------------------

Department of FRESHMAN ENGINEERING

Mathematics for Natural Sciences (Math 1007)

Test-I

Date: August, 2021 Time allowed: 45 min Maximum Score 15 %

Name Miracle Abrham Id No. ETS0901/13 Section Q1

Part I: Write "True" if the statement is correct and "False" if the statement is Wrong on the space provide (5 points)

True 1. If $p \Leftrightarrow q$ and $\neg p$ are $true,$ $then$ q is $false$

False 2. If $A \subseteq B^c$, then $A \cap B = \phi$.

False 3. If A$\subseteq$ B, then $A^c \subseteq B^c$

False 4. $(\exists x \in \Re)(\forall y \in \Re)(x^2 + yx = y + 1)$

True 5. Every composite number can be expressed as a product of its prime factors. and this factorization is unique except the order of the factors .

5

Part II: Short Answer Questions (5 points)
Write the Most simplified answer in the space provided

1. The GCF of two numbers is 3 and their LCM is 180. If one of the numbers is 45, then the second number is 12
2. Let A={ ϕ, a, c} and B= { {ϕ} ,c}, then $A \cap B$= C
3. Let $A \cup B = \{0,1,2,3,4,5\}$, $A \cap B' = \{0,1\}$ and $A \cap B = \{2,4\}$. Then $B \setminus A =$ ϕ P
4. The fraction form of $3.2\bar{5}$ is $293/990$
5. The argument of the complex numbers $z = \frac{3i}{-1-i}$ is $\tan^{-1}(1) = -\pi/4$ P